Introduction & Meaning of Statistics: Meaning of these words is "**Political State**" or a Government. Shakespeare used a word Statist is his drama Hamlet (1602). In the past, the statistics was used by rulers. The application of statistics was very limited but rulers and kings needed information about lands, agriculture, commerce, population of their states to assess their military potential, their wealth, taxation and other aspects of government.

Gottfried Achenwall used the word statistics at a German University in 1749 which means that political science of different countries. In 1771 W. Hooper (Englishman) used the word statistics in his translation of Elements of Universal Erudition written by Baron B.F Bieford, in his book statistics has been defined as the science that teaches us what is the political arrangement of all the modern states of th

The Word statistics have been derived from Latin word "**Status**" or the Italian word "**Statista** e known world. There is a big gap between the old statistics and the modern statistics, but old statistics also used as a part of the present statistics. During the 18th century the English writer have used the word statistics in their works, so statistics has developed gradually during last few centuries. A lot of work has been done in the end of the nineteenth century.

At the beginning of the 20th century, William S Gosset was developed the methods for decision making based on small set of data. During the 20th century several statistician are active in developing new methods, theories and application of statistics. Now these days the availability of electronics computers is certainly a major factor in the modern development of statistics.

Statistics is the science of data processing. It is a branch of knowledge dealing with collection, organization, presentation, analysis and interpretation of numerical data. Primarily statistics has two meanings. Statistics in plural sense refers to mass of information relating to characteristics of a population which can be defined as the aggregates of units under study. For example – the amount of recent per capita (average) expenditure of the official's workers of an industry, the annual birth rate or death rate of Bangladesh etc.

Statistics in singular sense refers to the science of statistics which performs statistical functions dealing with data collected for the purpose of specific study of a problem.

Definition of Statistics

The statisticians have defined statistics in different ways According to Prof. A.L. Bowley gives the three definitions of Statistics which as follows:

1. Statistics may be called the science of counting. This definition is too narrow because it covers only one aspect of the science, namely collection of data.
2. Statistics may be called the science of averages. This definition also is not satisfactory because averages are not only one of the devices used in statistical analysis.
3. Statistics is the science of the measurement of social organism, regarded as a whole in all its manifestations. This definition also is not satisfactory because it confines the scope of statistics only to sociology.

Boddington defines statistics as “the science of estimates and probabilities.” This definition also is unacceptable because estimates and probabilities are only a part of statistical method.

Croxton and Cowden say, “Statistics may be defined as a science of collection, presentation, analysis and interpretation of numerical data.” This definition clearly points out four stages in statistical investigation, namely: (1) collection of data (2) presentation of data (3) analysis of data and (4) interpretation of data.

However, to the above stages one more stage may be added and that is the organization of data. Thus, **statistics may be defined as the science of collection, organization, presentation, analysis and interpretation of numerical data.**

Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data. In applying statistics to, e.g., a scientific, industrial, or social problem, it is conventional to begin with a statistical population or a statistical model process to be studied. Populations can be diverse topics such as "all people living in a country" or "every atom composing a crystal". Statistics deals with all aspects of data including the planning of data collection in terms of the design of surveys and experiments.

Statistics is a form of mathematical analysis that uses quantified models, representations and synopses for a given set of experimental data or real-life studies. Statistics studies methodologies to gather, review, analyze and draw conclusions from data. Some statistical measures include central value measurement, regression analysis, variance analysis, time series analysis, probability measurement, sampling, hypothesis testing etc.

Statistics is the mathematical science involved in the application of quantitative principles to the collection, analysis, and presentation of numerical data. The practice of statistics utilizes data from some population in order to describe it meaningfully, to draw conclusions from it, and make informed decisions. The population may be a community, an organization, a production line, a service counter, or a phenomenon such as the weather. Statisticians determine which quantitative model is correct for a given type of problem and they decide what kinds of data should be collected and examined. Applied statistics concerns the application of the general methodology to particular problems. This often calls for use of the techniques of computer-based data analysis.

Statistical Method:

Statistical methods are the methods of collecting, summarizing, analyzing, and interpreting variable numerical data. Statistical methods can be contrasted with deterministic methods, which are appropriate where observations are exactly reproducible or are assumed to be so. While statistical methods are widely used in the life sciences, in economics, and in agricultural science, they also have an important role in the physical sciences in the study of measurement errors, of random phenomena such as radioactivity or meteorological events, and in obtaining approximate results where deterministic solutions are hard to apply.

Data collection involves deciding what to observe in order to obtain information relevant to the questions whose answers are required, and then making the observations. Sampling involves choice of a sufficient number of observations representing an appropriate population. Experiments with variable outcomes should be conducted according to principles of experimental design. ***Data summarization*** is the calculation of appropriate statistics and the display of such information in the form of tables, graphs, or charts. Data may also be adjusted to make different samples more comparable, using ratios, compensating factors, etc.

Statistical analysis relates observed statistical data to theoretical models, such as probability distributions or models used in regression analysis. By estimating parameters in the proposed model and testing hypotheses about rival models, one can assess the value of the information collected and the extent to which the information can be applied to similar situations. *Statistical prediction* is the application of the model thought to be most appropriate, using the estimated values of the parameters.

Finally we can say that **statistical methods are mathematical formulas, models, and techniques** that are used in statistical analysis of raw research data. The application of statistical methods extracts information from research data and provides different ways to assess the robustness of research outputs.

Objective of Statistics:

The main purpose of statistics is the collection of data regarding different aspects of life, presentation, analysis and coming to the conclusion with a reasonable interpretation. In this respect, the objectives of statistics are as follows:

(1).Statistics collects the numerical and analytical data regarding various aspects of our daily life.
(2).It presents the collected data attractively and simplifies them. For this classification, tabulation and graph-diagram are used.
(3).Analyzing the mathematical data it presents them to us in compact form.
(4).It deals with the characteristics of obtained information and compares among them.
(5).It deals with various methods of experiment.

Functions or Uses of Statistics

Statistics helps in providing a better understanding and exact description of a phenomenon of nature.

(1) Statistical helps in proper and efficient planning of a statistical inquiry in any field of study.
(2) Statistical helps in collecting an appropriate quantitative data.
(3) Statistics helps in presenting complex data in a suitable tabular, diagrammatic and graphic form for an easy and clear comprehension of the data.

(4) Statistics helps in understanding the nature and pattern of variability of a phenomenon through quantitative observations.

(5) Statistics helps in drawing valid inference, along with a measure of their reliability about the population parameters from the sample data.

(6) Statistics helps in providing a better understanding and exact description of a phenomenon of nature.

(7) Statistical helps in proper and efficient planning of a statistical inquiry in any field of study.

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(9) Statistics helps in providing a better understanding and exact description of a phenomenon of nature.

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(12) Statistics helps in presenting complex data in a suitable tabular, diagrammatic and graphic form for an easy and clear comprehension of the data.

(13) Statistics helps in understanding the nature and pattern of variability of a phenomenon through quantitative observations.

(14) Statistics helps in drawing valid inference, along with a measure of their reliability about the population parameters from the sample data.

Types of Statistics:

(i) Descriptive Statistics

(ii) Inferential or Inductive Statistics

Descriptive Statistics & Inferential or Inductive Statistics:

Descriptive & Inductive Statistics refer to the application of statistical methods & theories. Descriptive Statistics deals with collection, presentation, analysis & interpretation of statistical data without involving generalizations. On the other hand, generalization, predictions, estimations & decisions made from collected & analyzed data using statistical methods & theories come within the fold of Inductive Statistics. When we compute average with the sole purpose of describing a given phenomenon, we are using the mean as a tool of descriptive statistics, whereas if we make use of the average for making prediction or general conclusion, we are using it in inductive sense.

For example: Average salary of the employees of a factory may be described by calculating the mean salary of all the employees of the factory. This involves only description of mean salary. But if this mean averages of a particular factory is used as an estimate of average salary of the employees of all the factories in an industry instead of computing the average for all the employees in the industry then we use the mean in inductive sense.

Scope and Importance of Statistics:

1. ***Statistics and planning:*** Statistics in indispensable into planning in the modern age which is termed as "the age of planning". Almost all over the world the govt. are re-storing to planning for economic development.
2. ***Statistics and mathematics:*** Statistics are intimately related recent advancements in statistical technique are the outcome of wide applications of mathematics.
3. ***Statistics and modern science:*** In medical science the statistical tools for collection, presentation and analysis of observed facts relating to causes and incidence of dieses and the result of application various drugs and medicine are of great importance.
4. ***Statistics and* communication *engineering*** : In **communication engineering** there are several instances where **statistical** modeling is **used** for **system** planning, design, operation, and management. ... **Statistical** analysis and modeling of a network gives us information about the network deployment costs, network dynamics, and parameters of network traffic.
5. ***Statistics and* computer science:** **Statistics** is **used** for data mining, speech recognition, vision and image analysis, data compression, artificial intelligence, and network and traffic modeling. A **statistical** background is essential for understanding algorithms and **statistical** properties that form the backbone of **computer science**.

6. ***Statistics and economics:*** Statistical data and techniques of statistical analysis have to immensely useful involving economical problem. Such as wages, price, time series analysis, demand analysis.

7. ***Statistics and business:*** Statistics is an irresponsible tool of production control. Business executive are relying more and more on statistical techniques for studying the much and desire of the valued customers.

i. ***Marketing*****:** Statistical analysis is frequently used in providing information for making decision in the field of marketing. It is necessary first to find out what can be sold and the to evolve suitable strategy, so that the goods can be taken by the ultimate consumer. A skill full analysis of data on production purchasing power, man power, habits of competitors, habits of consumer, transportation cost should be considered to take any attempt to establish a new market.

ii. ***Production*****:** In the field of production statistical data and method play a very important role. The decision about what to produce? How to produce? When to produce? For whom to produce is based largely on statistical analysis.

iii. ***Finance:*** The financial organization discharging their finance function effectively depend very heavily on statistical analysis of peat and tigers.

iv. ***Banking*****:** Banking institute have found if increasingly to establish research department within their organization for the purpose of gathering and analysis information, not only regarding their own business but also regarding general economic situation and every segment of business in which they may have interest.

v. *Investment*: Statistics greatly assists investors in making clear and valued judgment in his investment decision in selecting securities which are safe and have the best prospects of yielding a good income.

vi. *Purchase:* The purchase department in discharging their function makes use of statistical data to frame suitable purchase policies such as what to buy? What quantity to buy? What time to buy? Where to buy? Whom to buy?

vii. *Accounting:* Statistical data are also employer in accounting particularly in auditing function, the technique of sampling and destination is frequently used.

viii. *Control:* The management control process combines statistical and accounting method in making the overall budget for the coming year including sales, materials, labor and other costs and net profits and capital requirement.

8. *Statistics and industry:* In industry statistics is widely used inequality control. In production engineering to find out whether the product is confirming to the specifications or not. Statistical tools, such as inspection plan, control chart etc.

9. *Statistics and psychology & education:* In education and physiology statistics has found wide application such as, determining or to determine the reliability and validity to a test, factor analysis etc.

10. Statistics *and war:* In war the theory of decision function can be a great assistance to the military and personal to plan "maximum destruction with minimum effort."

Limitations of Statistics

Statistics is indispensable to almost all sciences - social, physical and natural. It is very often used in most of the spheres of human activity. In spite of the wide scope of the subject it has certain limitations. Some important limitations of statistics are the following:

1. **Statistics does not study qualitative phenomena:** Statistics deals with facts and figures. So the quality aspect of a variable or the subjective phenomenon falls out of the scope of statistics. For example, qualities like beauty, honesty, intelligence etc. cannot be numerically expressed. So these characteristics cannot be examined statistically. This limits the scope of the subject.

2. **Statistical laws are not exact:** Statistical laws are not exact as incase of natural sciences. These laws are true only on average. They hold good under certain conditions. They cannot be universally applied. So statistics has less practical utility.

3. **Statistics does not study individuals:** Statistics deals with aggregate of facts. Single or isolated figures are not statistics. This is considered to be a major handicap of statistics.

4. **Statistics can be misused:** Statistics is mostly a tool of analysis. Statistical techniques are used to analyze and interpret the collected information in an enquiry. As it is, statistics does not prove or disprove anything. It is just a means to an end. Statements supported by statistics are more appealing and are commonly believed. For this, statistics is often misused. Statistical methods rightly used are beneficial but if misused these become harmful. Statistical methods used by less expert hands will lead to inaccurate results. Here the fault does not lie with the subject of statistics but with the person who makes wrong use of it.

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Questions

1. What is statistics? Discuss the importance of statistics in Computer science.
2. Discuss applications and limitations of statistics.
3. Explain some important functions of statistics.
4. How does statistics help a programmer?
5. ‘Statistics is the science of average’ – Explain.
6. What is the basic difference between Descriptive and Inferential Statistics?